Software Architecture

N-Tier / Layerd Architecture

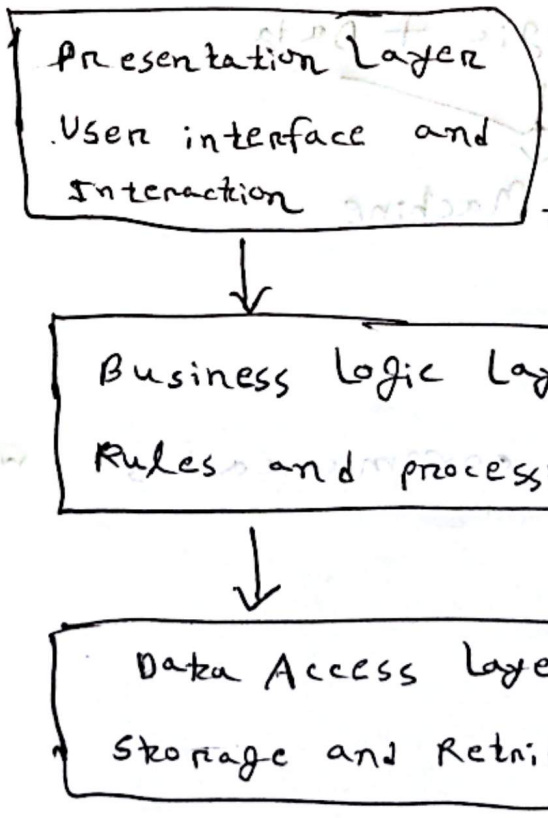
N-Tier Architecture ৭ system কে বিভিন্ন layer/tier ৭ ভাগ করা হয়।

Organize a system into layers or tiers, each with a distinct responsibility.

প্রতিটি Layer ৫র কাজ আলাদা।

Basic Layerd Structure

Ex: Login form
Buttons
web page
Mobile UI



User যে interface দেখে এবং যেটার মা system এর সাথে interact করে।

Ex: User login করলে password ঠিক আছে কিনা check করা

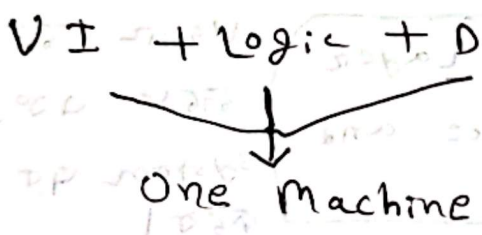
N-Tier Architecture

An approach where a project or program is distributed among N seperate computers in a distributed network.

Types of N-Tier Architecture

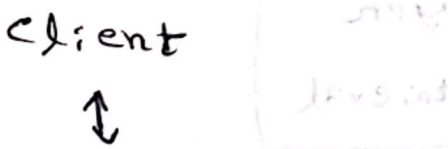
1 - Tier

Interface, Logic এবং data একই machine / application এ থাকে।



2 - Tier

Client directly communicates with the database server.



3- Tier

Client, application server, database server আলাদা

Client Tier
↓
Application Tier
↓
Database Tier

1. Client Tier (Browser/ Desktop / mobile)

User এখান থেকে system ব্যবহার করে।

2. Application Tier (Business Logic server)

এখানে Application - এর main processing হয়।

Ex:
- Order Validation
- Total calculation
- Business Rules

3. Database Tier

- Products
- Users
- Orders
- Transactions

Presentation Tier

Presents the application interface and translates tasks/results into a form users can understand.

Logic Tier

Coordinates the application, processes commands, makes logical decisions and performs calculations.

Data Tier

Stores and retrieves information from databases or file systems.

2 - Tier Architecture

Client Tier ⟷ Database Server

client সরাসরি Database server এর সাথে communicat করে।

The client sends queries directly to the databa server.

1 - Tier Architecture

1 - Tier Architecture means the whole system runs on one computer / one tier.

The following parts are located together:

- User Interface
- Business Logic
- Local Data

Ex:

একটি personal Expense app:

Expense input দিলে → application calculation করবে → local storage এ save করবে।

সব একটি computer এ হবে।

1 - Tier = Everything on one Machine.

Software Design Software Architecture

Impacts of Multi Tiering

দুই ধরনের quality আলোচনা করেছে।

1. Operational Qualities
2. Development Qualities.

Operational Qualities

1. Scalability: প্রয়োজন হলে একটি tier আলাদাভাবে scale করা যায়।

2. Reliability:

প্রত্যেক অংশের responsibility আলাদা হওয়ায় system বেশি organized থাকে।

3. Fault Tolerance

কোন একটি অংশে সমস্যা হলে সেটিকে আলাদা করে handle/recover করা সহজ হয়।

4. Security

Sensitive data বা tier-এ direct access বন্ধ বা সীমিত করা যায়।

Development Qualities

Maintainability

একটি নির্দিষ্ট tier এর পরিবর্তন সেই tier-এর মধ্যেই সীমাবদ্ধ রাখা সহজ।

Reusability

Shared logic/services একাধিক client ব্যবহার করতে পারে।

Software Architectural Patterns

A reusable high-level structure for organizing components and their interactions.

MVC

It's a 3-Tier Architecture.

Advantages

- Rapid Development
- Low coupling (Dependency কম)
- Efficient debugging
- High maintainability

Ex:

Moderate to large scale project.

উদাহরণ, frontend এর একটি কাজ থাকবে, controlling এর ও থাকবে, database এর ও থাকবে।

MVC Components

(i) Model

Model মূলত application এর data এবং data related logic handle করে।

(ii) View

The view contains the UI logic of the application. It displays information to the user

(iii) controller

controller হলো model এবং view এর মাঝখানের coordinator.

Monolithic Architecture

ఇది 1 – Tier Architecture.

Ex: A university portal packages login, registrati payment, and results into one application deployed together.

Advantages

- Low cost architecture
- Simple for development and deployment
- Difficult for maintenance and scalability as the system grows.
- Tight coupling

Ex: Airbnb, cash counter software

Microservice Architecture

এটিও একটি 3-Tier Architecture.

কিন্তু প্রতিটা ~~server~~ service এর জন্য 3-Tier Architecture Build করা হয়।

Ex: Pathao App (Food, Care, Bike)

এটা কিছুটা costly, but যারা afford করতে পারবে, তারা এই microservice use করবে easily.

Advantages

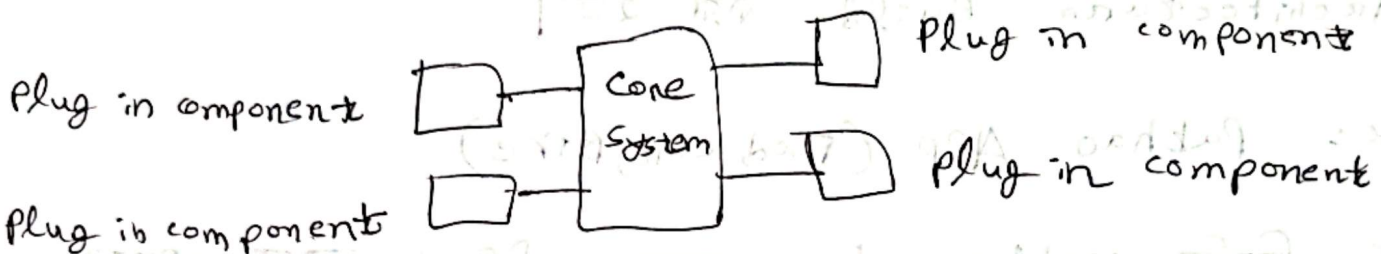
- ~~small~~ Breaks it down into a collection of smaller independent unit.
- Easy maintainability, Scalability, low coupling
- Independent Development and Deployment
- Specific Function per Service.

Disadvantages

- Extra complexity
- Higher costs.
- Cross-cutting concerns.

Micro-Kernel Architecture Pattern

একটি core function থাকে, যার সাথে কিছু plugins / addons দিয়ে নতুন feature বানানো যায়।



Ex: VS code / Intellij Idea

Advantages

- Easy to extend
- Core system remains small
- Flexible
- Specialized Functionality

Disadvantages

- Plug-in integration can be complex
- Dependency on plug-in compatibility
- Managing many plug-ins can become difficult.

Event-Driven Architecture

- প্রতিটি Event / occurance এর উপরের Architecture হয়।

Event হইলে, তাহলে এই Architecture Run করবে এইভাবে কাজ করে।

Ex: Ride Sharing App, Ecommerce, Online Order

Advantages

- Fast Development
- Easy Debugging
- High Performance
- Low coupling

Ex: Food Order

